Module 4 Assignment

Investing in Nashville

ALY 6020: Predictive Analytics

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Instructor: Professor Justin Grosz

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Author:

Brian Griffin

Introduction

The real estate market has been competitive across the United States over recent years due to low interest rates giving investors greater ability to afford homes. This environment, along with limited supply, has allowed home prices to reach record highs across the nation (Rothstein, R., 2023). Given this environment, along with a recent shift towards higher interest rates made by the Federal Reserve to curb inflation, it is difficult to find value in a real estate investment since prices have been inflated to such high levels and financing has become more difficult. In this week's case study, a real estate company is considering making a large investment in the Nashville, Tennessee area. The company will be making its investment decision based on 'Sale Price Compared to Value', in an effort to find properties that are undervalued. According to Zillow, Nashville is predicted to be the fifth hottest real estate market of 2023 driven by factors such as home value growth, housing market velocity, home construction activity, and a relatively fast-moving inventory (Kennedy, P., 2023). To understand the most important variables in making an investment decision in this growing market, I will be conducting multiple machine learning models based on a recent housing dataset provided covering the Nashville area. These models include logistic regression, decision tree, random forest, and gradient boost. These models will be built using Python programming, and benchmarking will be performed to identify the best model to assist the company in discovering undervalued properties in Nashville.

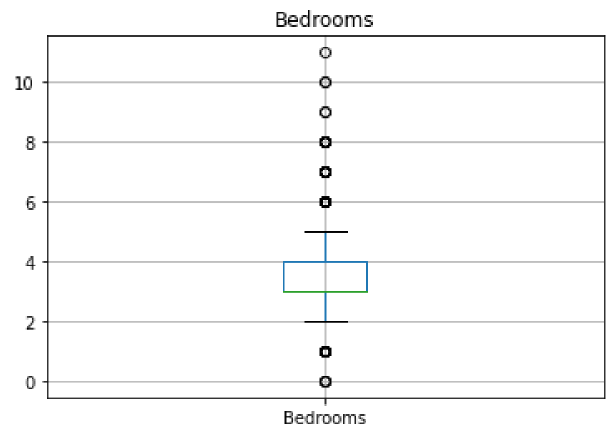
Data Cleansing

The dataset provided for analysis contains 22,651 records of individual property sales consisting of 26 variables, both quantitative and qualitative, that describe the property, its location, and primary characteristics. The dataset also includes a variable described as "Sale

Price Compared to Value", which indicates if the sale was Over or Under valued. Extensive data cleaning was performed to ensure the dataset was ready for additional analysis and modeling. The 'Suite/ Condo #' variable was removed because it does not contain any data. Multiple nulls were identified and addressed: 2 records without 'Property Address' or 'Property City' values were removed from the dataset as the information for those records did not seem reliable if location information could not be found. Additionally, 3 records without any 'Bedrooms' populated were removed as this is likely a critical value in determining the value of a house. One record was missing data for 'Finished Area', which was filled with the median value of that column rather than mean, as outliers were skewing the average. There was also a record missing 'Foundation Type', which was replaced with the value of 'CRAWL', a value that represents 63% of the overall dataset. Another record was missing a value for 'Full Bath'. This particular record was a 3 bedroom house, and the mean 'Full Bath's for a 3 bedroom house is 2, therefore the missing value was replaced with 2 'Full Bath's. Finally, 108 records returned null values for 'Half Bath'. The median amount of 'Full Bath's' for these particular records is 2, therefore I assumed that these houses did not likely contain any 'Half Bath's' within the property. I replaced these null values with 0.

Additional analysis was performed to determine if any unrealistic values were found in the dataset that may needed to be removed. 167 properties appeared to be outliers based on 'Year Built' as they were built before 1901. After analyzing these records, they appeared to be real data points as it is possible for a house to be built during that period, although unlikely, and they were kept in the dataset. There was also an outlier that contained 17 acres of land within the 'Acreage' variable, but this also seemed like a realistic record based on the other data associated with it and was kept in the dataset. There were outliers found within the 'Land Value' and

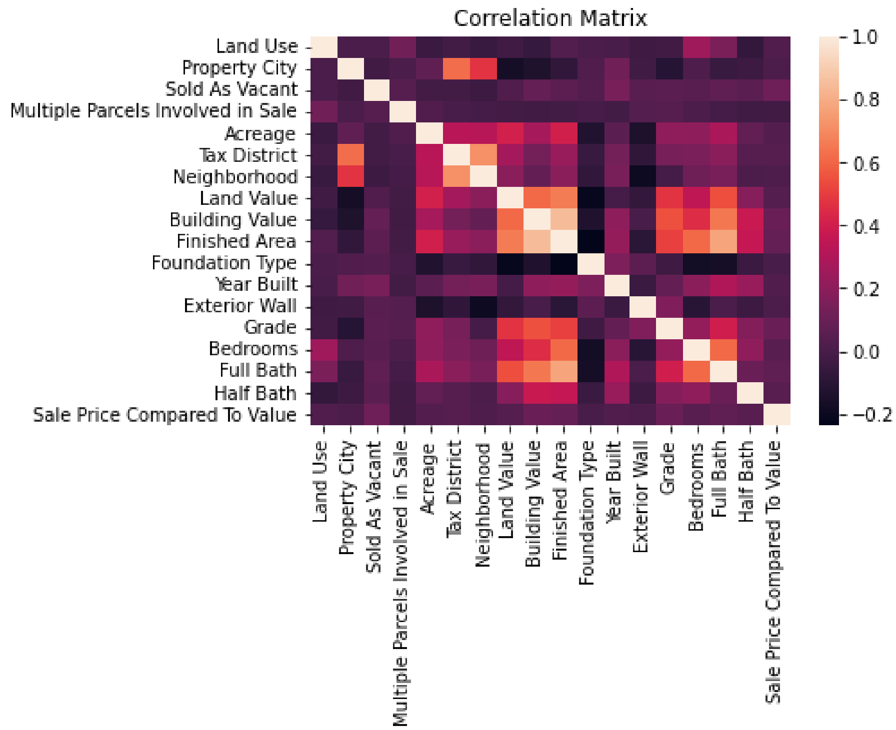
‘Building Value’ attributes with regards to high values, but these properties had characteristics (such as multiple bedrooms, large finished area) that appeared to make these values seem like valid records. The only outliers that were removed from the dataframe were found when analyzing the ‘Bedrooms’ boxplot.



By analyzing the visualization above, I was able to identify 3 records within the dataset that contained zero bedrooms. After analyzing these records, they appeared to be unusual properties based on characteristics and had very low ‘building value’ amounts. I did not think that including these records in the dataset would assist in analysis since it is unlikely for a buyer to desire a property without any bedrooms, so I removed them from the population. Additionally, I dropped the following columns from the dataset before performing any modeling: 'Unnamed: 0', 'Parcel ID', 'Property Address', 'Sale Date', 'Legal Reference', ‘City’, ‘State’. Although some of these variables are valuable for identification purposes (such as 'Parcel ID', 'Property Address', ‘City’, ‘State’), I did not think they would be valuable for modeling. For example, the ‘City’ variable is essentially identical to the ‘Property City’ variable, and all properties are the same within the ‘State’ variable (TN). After the completion of data modeling, 22,642 records consisting of 18 attributes remain within the dataframe.

Exploratory Data Analysis (EDA)

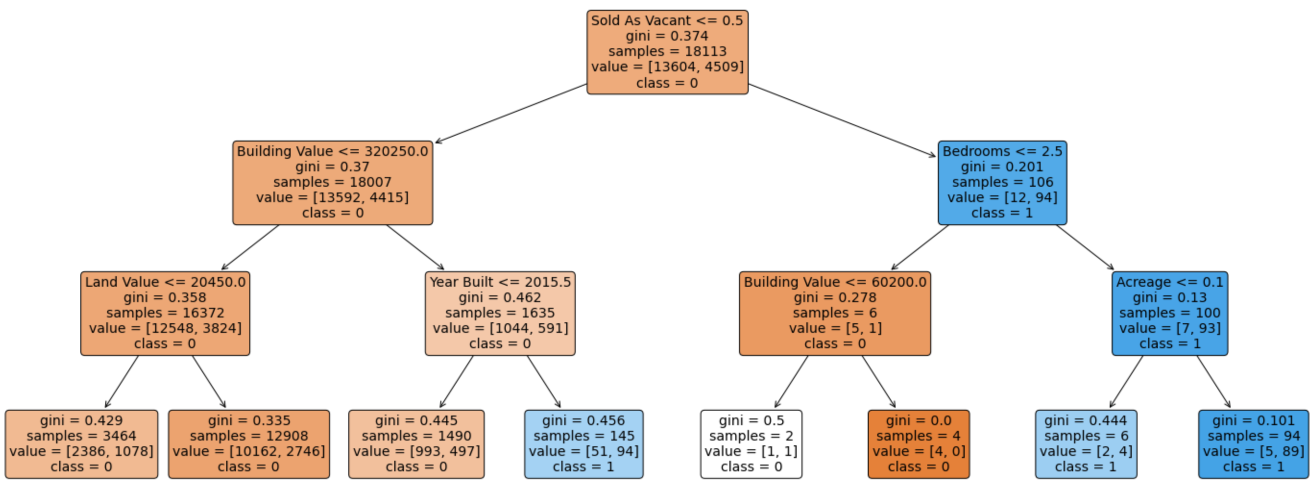
Exploratory data analysis (EDA) was conducted to find any patterns and additional insights into the data. Before removing 'Parcel ID' from the dataset, it was interesting to find that 2,930 of the properties in the dataset had been resold (based on unique 'Parcel ID's in the dataset). Depending on sale price, they may have different valuation ('Over' or 'Under') with identical characteristics. I also created a separate dataframe of all properties that were deemed to be undervalued based on 'Sale Price Compared to Value' to determine if they had any unique characteristics compared to the full dataset. 25% of all sales within the total dataset were identified as undervalued. Of these undervalued records, 2.1% more of this sample was 'Sold As Vacant' compared to the overall dataset. Additionally, I found that the undervalued properties were skewed towards higher 'Grade' compared to the larger dataset. 20% of the undervalued properties were identified as 'Grade' A or B compared to only 15% of the total dataset. Both 'Sold As Vacant' and 'Grade' are attributes that should be considered when performing modeling. I found that many of the quantitative variables associated with undervalued properties were similar in terms of central tendency compared to the larger dataset, which makes it difficult to find any obvious themes before performing modeling. Before conducting modeling, I created a correlation matrix to search for relationships between the variables. Shown in the figure at the top of page 5, if it is difficult to find any strong positive correlation between the dependent variable, 'Sale Price Compared to Value', and other variables in the dataset based on visualization alone. There is however correlation between some of the other variables, that may be a sign of multicollinearity that will need to be considered when performing modeling.



Modeling

Four models were built to determine the optimal combination of variables in determining 'Sale Price Compared to Value'. These models include logistic regression, decision tree, random forest, and gradient boost modeling. The sklearn package within Python was leveraged for all the models. The models incorporated a train / test split of 80% train data and 20% test data. After performing logistic regression, I identified the following variables as significant based on low P-Value's: 'Sold as Vacant', 'Tax District', 'Land Value', 'Building Value', 'Year Built', and 'Grade'. It is interesting that 'Sold as Vacant' and 'Grade' were identified as significant variables, as this was consistent with initial exploratory data analysis performed earlier. The logistic model yielded accuracy of 76%. However, despite this accuracy score, the model did not perform well in identifying undervalued properties based on the confusion matrix and classification report which is presented at the end of this section. This is a significant drawback since identifying undervalued properties is the primary goal of the real estate company to enable them to make an informed investment decision.

After logistic regression was performed, a decision tree was modeled to determine if it could better identify undervalued properties based on the dataset provided. I chose to use a max_depth of 3 when fine tuning the parameters of the model. I had initially used a max_depth of 5, but it made the model more complex, difficult to analyze, and did not provide any significant improvements in terms of accuracy or classification scores. Below is the visualization of the algorithm produced from my decision tree model:



Similar to logistic regression, the decision tree identified ‘Sold as Vacant’ as a key variable in determining ‘Sale Price Compared to Value’. The tree also incorporated variables such as ‘Building Value’, ‘Land Value’, and ‘Year Built’ as important decisions in the algorithm, similar to logistic regression. Conversely, the decision tree placed more importance with variables such as ‘Bedrooms’ and ‘Acreage’ as compared to the significance derived from the logistic model. The decision tree yielded slightly improved accuracy of 76.7% and performed better at predicting undervalued properties based on the classification report and confusion matrix.

The third model produced was a random forest model. A random forest is an ensemble model that builds multiple decision trees with the goal of determining a more accurate and stable prediction (Srivastava, T., 2020). After fine tuning the model, I decided to implement n_estimators at 35, which means the model uses a maximum of 35 trees before taking the maximum voting or averages of predictions. Using additional n_estimators did not have a significant impact on improving accuracy of the model, and performance began to reduce as the model ran slower. After producing a more fine-tuned model, I was only able to deliver with an overall accuracy of 72.4%. However, although overall accuracy was reduced, the model did a better job at reducing false negatives compared to the previous two models, which could potentially reduce cost for the real estate company implementing the model, since making incorrect predictions on undervalued properties that aren't actually undervalued could be very expensive.

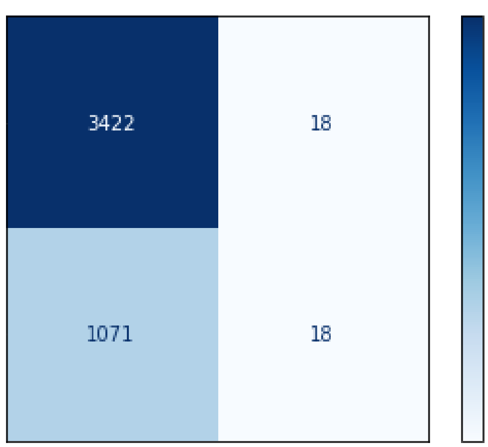
Finally, a gradient boost model was implemented. Another ensemble model, gradient boosting builds a new tree that is fit on a modified version of the original dataset. After evaluating the first tree, in which each observation is given an equal weight, weights are increased and decreased based on how difficult they are to classify. The model trains in a gradual, additive, and sequential manner (Singh, H., 2018). For my unique model, I performed fine tuning using both the n_estimators and learning_rate parameters. I landed on 40 n_estimators as the number of sequential trees to be modeled, which appeared to provide more accuracy than the 35 trees modeled using random forest. I also adjusted the learning_rate parameter, which controls the magnitude of change in the estimates, however I found that the default value of 0.1 was sufficient; altering the learning rate in my model did not provide additional accuracy sufficient to offset any reduction in performance. Overall, the gradient boost model yielded accuracy of 76.8%, performing well at predicting both undervalued and overvalued properties based on results from the confusion matrix and classification report. Overall results from the four models below are summarized below:

Classification Report					
Model	**Prediction**	**Precision**	**Recall**	**F1-Score**	**Support**
Logistic Regression	0 (Over Valued)	0.76	0.99	0.86	3440
	1 (Under Valued)	0.50	0.02	0.03	1089
Decision Tree	0 (Over Valued)	0.77	1.00	0.87	3440
	1 (Under Valued)	0.76	0.05	0.09	1089
Random Forest	0 (Over Valued)	0.77	0.90	0.83	3440
	1 (Under Valued)	0.35	0.17	0.23	1089
Gradient Boost	0 (Over Valued)	0.77	1.00	0.87	3440
	1 (Under Valued)	0.77	0.05	0.09	1089

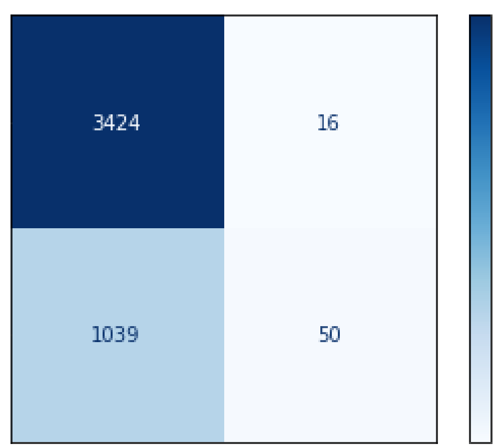
Confusion Matrix

Top Left Quadrant (**True Positive**)	Top Right Quadrant (**False Positives**)
Bottom Left Quadrant (**False Negatives**)	Bottom Right Quadrant (**True Negatives**)

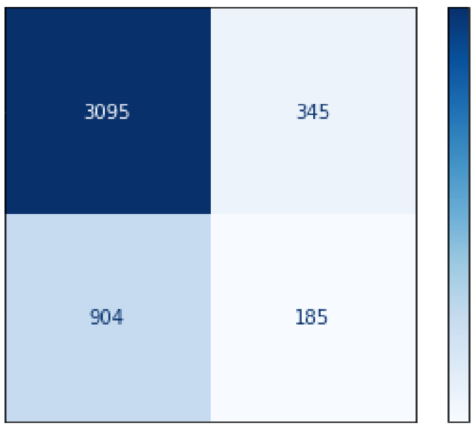
Logistic Regression



Decision Tree



Random Forest



Gradient Boosting

Conclusion

After completing modeling, benchmarking was performed to determine the best model in predicting ‘Sale Price Compared to Value’ in the Nashville real estate dataset. Benchmarking requires taking multiple variables into consideration, including accuracy, results from classification, results from the confusion matrix, and performance. In terms of overall accuracy, the decision tree and gradient boosting performed the highest, with 76.7% and 76.8% respectively. Logistic Regression and Random Forest yielded lower accuracy results, however Random Forest had the best performance in terms of optimizing the confusion matrix, reducing false negatives compared to the other models, when a case is positive but predicted negative. The random forest model also performed well in terms of recall, the percentage of correct positive predictions relative to total actual positives (Zach, 2022). Ultimately the logistic regression performed the worst in terms of benchmarking, specifically accuracy and reducing false negatives, although the model was helpful in determining what variables were statistically significant. Ultimately, my recommendation is that the real estate company choose the gradient boosting model. The gradient boosting model had strong overall accuracy, reduced false negatives, had strong classification results especially in terms of precision, the percentage of correct positive predictions relative to total positive predictions, for both over valued and under valued predictions. Additionally, the gradient boosting model did not suffer from performance results, as selecting 40 n_estimators did not inhibit computation of the modeling. The gradient boosting model is an effective tool in determining ‘Sale Price Compared to Value’.

References

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Appendix

Logistic Regression Results

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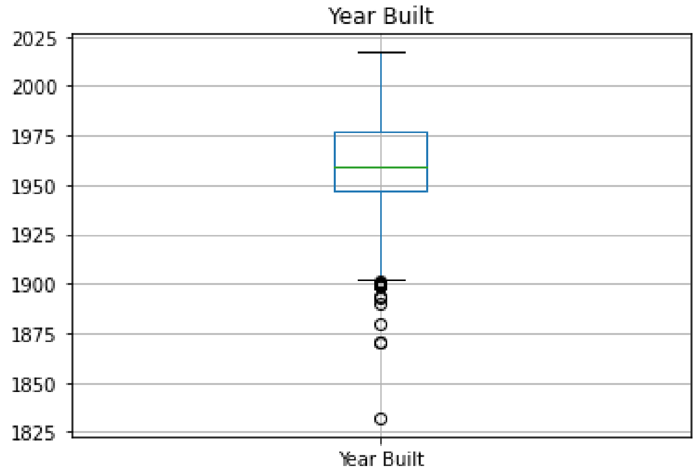
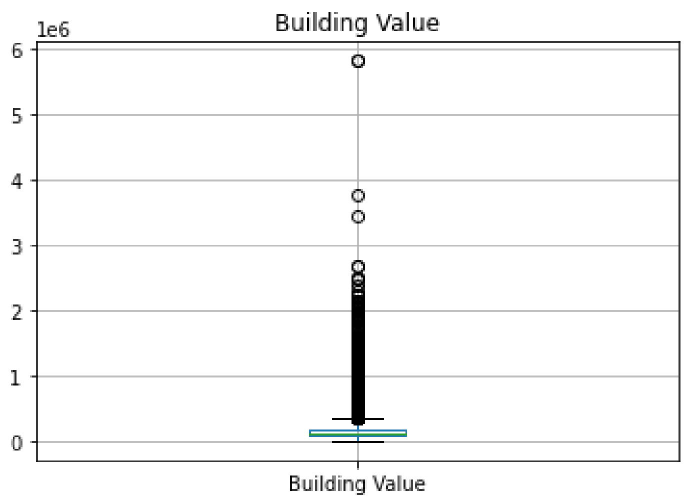
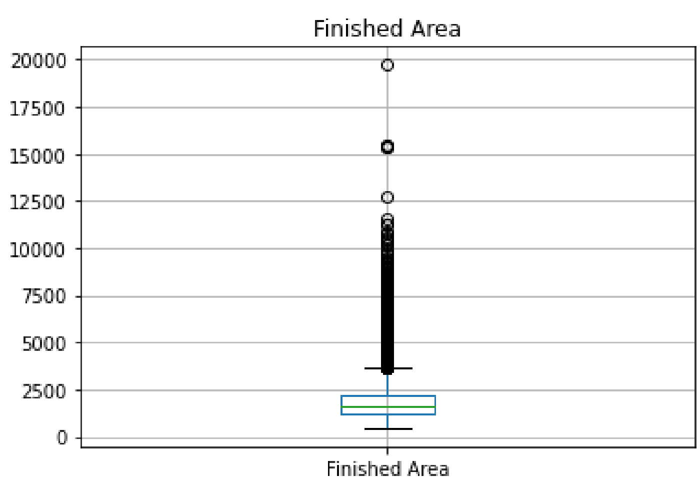
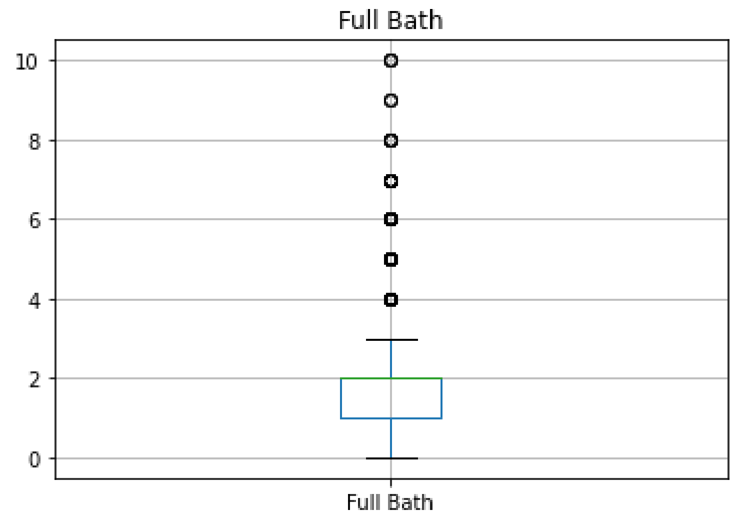
                           Logit Regression Results
==========================================================================================
Dep. Variable:     Sale Price Compared To Value   No. Observations:                18113
Model:                                    Logit   Df Residuals:                    18096
Method:                                     MLE   Df Model:                           16
Date:                          Sat, 04 Feb 2023   Pseudo R-squ.:                 0.02460
Time:                                  12:06:00   Log-Likelihood:                -9914.3
converged:                                 True   LL-Null:                       -10164.
Covariance Type:                      nonrobust   LLR p-value:                 3.173e-96
=====================================================================================================
                                    coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------------
Land Use                          0.2401      0.069      3.478      0.001       0.105       0.375
Property City                    -0.0114      0.016     -0.732      0.464      -0.042       0.019
Sold As Vacant                    3.2033      0.321      9.965      0.000       2.573       3.833
Multiple Parcels Involved in Sale -0.4048     0.135     -2.991      0.003      -0.670      -0.140
Acreage                           0.0630      0.032      1.944      0.052      -0.001       0.126
Tax District                      0.0923      0.024      3.868      0.000       0.046       0.139
Neighborhood                  -1.253e-05   1.23e-05     -1.018      0.309   -3.67e-05    1.16e-05
Land Value                    -1.597e-06   2.61e-07     -6.108      0.000   -2.11e-06   -1.08e-06
Building Value                 9.966e-07   1.98e-07      5.033      0.000    6.09e-07    1.38e-06
Finished Area                 -2.745e-05   4.27e-05     -0.642      0.521      -0.000    5.63e-05
Foundation Type                   0.0070      0.017      0.412      0.680      -0.026       0.041
Year Built                       -0.0006   5.29e-05    -12.229      0.000      -0.001      -0.001
Exterior Wall                    -0.0166      0.017     -0.999      0.318      -0.049       0.016
Grade                             0.1789      0.023      7.871      0.000       0.134       0.223
Bedrooms                         -0.0260      0.029     -0.892      0.372      -0.083       0.031
Full Bath                         0.0396      0.033      1.213      0.225      -0.024       0.104
Half Bath                         0.0444      0.040      1.104      0.269      -0.034       0.123
=====================================================================================================
```

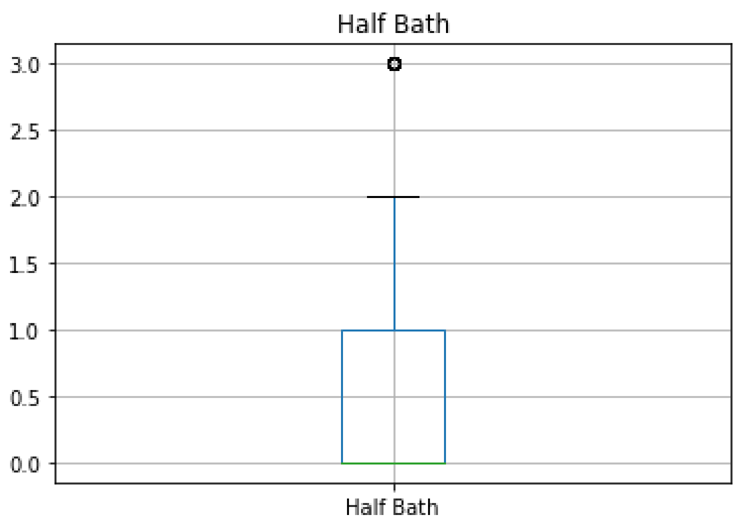
Boxplots (Identify Outliers)








Full Bath
10
8
6
4
2
0
Full Bath


Half Bath
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Half Bath

Histograms (Distribution)